

Appendix Table A4. Feature Importance

Rank	Feature	Importance mean	Importance std.	Window	Model
1	p1_gaze_pitch_max	0.0103	0.0016	5s	Random Forest
2	frame_gaze_valid_transition_coun	0.0102	0.0018	5s	Random Forest
3	frame_pose_valid_ratio	0.0094	0.0031	5s	Random Forest
4	frame_gaze_valid_count	0.0093	0.0030	5s	Random Forest
5	p1_task_gaze_count	0.0079	0.0014	5s	Random Forest
6	p2_gaze_yaw_min	0.0078	0.0034	5s	Random Forest
7	root_distance_max	0.0077	0.0031	5s	Random Forest
8	p1_task_gaze_angle_max	0.0076	0.0023	5s	Random Forest
9	frame_interaction_valid_count	0.0075	0.0028	5s	Random Forest
10	frame_pose_valid_count	0.0074	0.0025	5s	Random Forest
11	p2_gaze_pitch_max	0.0074	0.0020	5s	Random Forest
12	hand_to_hand_min_min	0.0056	0.0028	5s	Random Forest
13	frame_interaction_valid_ratio	0.0056	0.0030	5s	Random Forest
14	p2_motion_speed_std	0.0050	0.0017	5s	Random Forest
15	p1_gaze_pitch_min	0.0048	0.0008	5s	Random Forest

Top 15 permutation feature importances for the best condition classifier analysis. Note: Some top features are validity/visibility features, so the condition classifier should not be interpreted as purely behavioral without further validity-controlled analysis.